MODULE 3

Natural Disasters

"Yes, a dark time passed over this land, but now there is something like light."

— DAVE EGGERS

ESSENTIAL QUESTION

How can learning about natural disasters make us safer?

MAIN SELECTIONS

1	**Eruption! Volcanoes and the Science of Saving Lives**	· Elizabeth Rusch (photos Tom Uhlman)	NARRATIVE NONFICTION
2	**Between the Glacier and the Sea: The Alaska Earthquake**	· from the Valdez Museum	MEDIA: VIDEO
3	**Quaking Earth, Racing Waves**	· Rachel Young	INFORMATIONAL TEXT
4	**Hurricanes: The Science Behind Killer Storms**	· Alvin & Virginia Silverstein and Laura Silverstein Nunn	INFORMATIONAL TEXT

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B · Reading Comprehension (32) ..... p. 6

C · IB Critical Thinking (5) ..... p. 12

D · MAP Practice (20) ..... p. 15

A Worksheet A — Vocabulary

Module 3 · Natural Disasters · Grade 5 · 31 Items

CCSS L.5.4 · L.5.5 · L.5.6

NAME _____ DATE _____

Directions. Read each question carefully. Watch out for tricky choices. Fill in the circle next to the BEST answer.

BIG IDEA WORDS

Q1 L.5.6

notable

If something is **notable**, it is —

- ◯ **A.** too small to see.
- ◯ **B.** printed on a map.
- ◯ **C.** worth noticing.
- ◯ **D.** kept completely secret.

Q2 L.5.6

tremor

A **tremor** is —

- ◯ **A.** a tall ocean wave.
- ◯ **B.** a warm summer wind.
- ◯ **C.** a loud clap of thunder.
- ◯ **D.** a small earthquake or uncontrolled shaking in a body part.

Q3 L.5.5 TRAP

spontaneous

A **spontaneous** action is one that —

- ◯ **A.** happens naturally and isn't planned.
- ◯ **B.** is planned weeks ahead of time.
- ◯ **C.** follows a careful written schedule.
- ◯ **D.** takes a very long time to finish.

Q4 L.5.6

hazard

A **hazard** is —

- ◯ **A.** a safe and quiet place.
- ◯ **B.** a helpful piece of advice.
- ◯ **C.** a danger.
- ◯ **D.** a kind of measuring tool.

CRITICAL VOCABULARY · ERUPTION!

Q5 L.5.4

seismographs

The **seismographs** recorded the first earthquake. **Seismographs** are —

- ◯ **A.** trucks that carry people out of danger.
- ◯ **B.** maps that show roads near a volcano.
- ◯ **C.** photographs taken from an airplane.
- ◯ **D.** instruments that measure and record details about earthquakes, such as their strength and how long they last.

Q6 L.5.4

evacuation

Families moved to **evacuation** camps. An **evacuation** is —

- ◯ **A.** the act of building a new road.
- ◯ **B.** the act of moving from a dangerous area to a safer one.
- ◯ **C.** the act of repairing a broken machine.
- ◯ **D.** the act of counting how many people are in a town.

Q7 L.5.4

reservoir

Magma collected in a **reservoir** under the volcano. A **reservoir** is —

- ◯ **A.** a sudden burst of fire.
- ◯ **B.** a path down a mountain.
- ◯ **C.** a place where a supply of something is collected.
- ◯ **D.** a warning sign posted on a fence.

Q8 L.5.4

conferring

The scientists were **conferring** before they decided. If you are **conferring** with someone, you are —

- **A.** arguing with no purpose.
- **B.** discussing an idea or trying to make a decision.
- **C.** ignoring what they say.
- **D.** drawing a picture of them.

Q9 L.5.5 TRAP

consequences

The **consequences** weighed on their minds. **Consequences** are —

- **A.** the outcomes or effects of events.
- **B.** the causes that come before events.
- **C.** the names given to large storms.
- **D.** the tools used to dig in the ground.

CRITICAL VOCABULARY · ERUPTION! (CONTINUED)

Q10 L.5.4

widespread

The volcano caused **widespread** devastation. If something is **widespread**, it —

- **A.** happens over a large area or among many people.
- **B.** happens in only one small spot.
- **C.** happens deep underground.
- **D.** happens only at night.

Q11 L.5.5 TRAP

alarming

The instruments made **alarming** marks. Something that is **alarming** —

- **A.** makes you feel calm and safe.
- **B.** makes you feel sleepy and bored.
- **C.** makes you worry that something bad may happen.
- **D.** makes you laugh out loud.

Q12 L.5.4

victim

The scientists did not want to fall **victim** to an eruption. If you fall **victim** to something, you —

- **A.** teach other people about it.
- **B.** safely escape from it.
- **C.** measure it with a tool.
- **D.** suffer or die because of it.

CRITICAL VOCABULARY · BETWEEN THE GLACIER AND THE SEA

Q13 L.5.5 TRAP

prior

The town was calm in the days **prior** to the quake. **Prior** means —

- **A.** coming after everything else.
- **B.** happening far away.
- **C.** staying exactly the same.
- **D.** coming before, in time or order.

Q14 L.5.4

magnitude

The quake's **magnitude** was the largest ever recorded in North America. In geology, **magnitude** refers to —

- **A.** the color of the rocks.
- **B.** the rank of an earthquake; its greatness of size.
- **C.** the time of day it happened.
- **D.** the name of the town nearby.

Q15 L.5.5 TRAP

literally

The ground **literally** split apart. If you say something **literally** happened, it means —

- **A.** it actually happened, and you aren't exaggerating or using a metaphor.
- **B.** it was only a make-believe story.
- **C.** it was an exaggeration to sound exciting.
- **D.** it might happen someday.

Q16 L.5.4

maintenance

The oil tanks needed regular **maintenance**. The **maintenance** of something is —

- **A.** the act of caring for it and repairing it when needed.
- **B.** the act of selling it to someone.
- **C.** the act of hiding it away.
- **D.** the act of measuring its size.

CRITICAL VOCABULARY · QUAKING EARTH, RACING WAVES

Q17 L.5.4

parallel

The two cracks ran **parallel** to each other. If two things are **parallel**, they —

- **A.** cross each other in the middle.
- **B.** move in the same direction and never touch or intersect.
- **C.** spin around in fast circles.
- **D.** melt together into one piece.

Q18 L.5.4

lateral

The plates made a **lateral** movement. If something moves in a **lateral** way, it moves —

- **A.** straight up into the sky.
- **B.** deep into the ground.
- **C.** side to side.
- **D.** in a slow circle.

CRITICAL VOCABULARY · QUAKING EARTH, RACING WAVES (CONTINUED)

Q19 L.5.5 TRAP

destruction

The wave caused great **destruction**. **Destruction** is —

- **A.** the act of building something new.
- **B.** the act of cleaning a place.
- **C.** the act of destroying or ruining something.
- **D.** the act of guarding a town.

Q20 L.5.4

triggered

The earthquake **triggered** a tsunami. If you **triggered** an action or event, you —

- **A.** did something to start it.
- **B.** tried to stop it from happening.
- **C.** slowly watched it from far away.
- **D.** wrote a report about it later.

Q21 L.5.4

thrust

The moving rock can **thrust** the water upward. To **thrust** is to —

- **A.** push something with great force.
- **B.** pull something gently closer.
- **C.** hold something perfectly still.
- **D.** measure something carefully.

Q22 L.5.4

radiate

The waves **radiate** out from the center. To **radiate** is to —

- **A.** sink straight down.
- **B.** freeze into solid ice.
- **C.** stop without warning.
- **D.** spread out in waves or rays.

Q23 L.5.5 TRAP

modified

The town built a **modified** plan after the tsunami. A **modified** version of something is —

- **A.** an exact copy with nothing different.
- **B.** the very first version ever made.
- **C.** a broken version that no longer works.
- **D.** a revised, or changed, version.

CRITICAL VOCABULARY · HURRICANES

Q24 L.5.4

surge

A storm **surge** pushed the sea onto the land. A **surge** is —

- ◯ **A.** a slow, steady drip.
- ◯ **B.** a sudden powerful movement forward or upward.
- ◯ **C.** a quiet, still pool.
- ◯ **D.** a thin layer of dust.

CRITICAL VOCABULARY · HURRICANES (CONTINUED)

Q25 L.5.5 TRAP

eventually

A hurricane grows, but **eventually** it dies out. To say something will happen **eventually** means —

- ◯ **A.** it will happen at this very second.
- ◯ **B.** it will never happen at all.
- ◯ **C.** it will happen at some time, usually after a series of other events.
- ◯ **D.** it already happened long ago.

Q26 L.5.4

regions

More hurricanes hit some **regions** than others. **Regions** are —

- ◯ **A.** kinds of weather.
- ◯ **B.** areas.
- ◯ **C.** seasons of the year.
- ◯ **D.** kinds of boats.

Q27 L.5.4

mobile

Many of the buildings were **mobile** homes. Something that is **mobile** is —

- ◯ **A.** able to move or be moved easily.
- ◯ **B.** built deep underground.
- ◯ **C.** made entirely of glass.
- ◯ **D.** painted a bright color.

Q28 L.5.5 TRAP

anchored

The houses were not well **anchored** to the ground. Something that is **anchored** is —

- **A.** loose and free to roll away.
- **B.** firmly attached to something else or weighed down so it won't move easily.
- **C.** floating high in the air.
- **D.** spinning in fast circles.

Q29 L.5.4

foundations

Floodwater washed away the soil around the **foundations** of houses. **Foundations** are —

- **A.** the roofs at the very top of houses.
- **B.** the windows along the sides of houses.
- **C.** the fences that surround houses.
- **D.** the base pieces houses and buildings are built upon.

Q30 L.5.4

category

A scale ranks storms from **Category** 1 to Category 5. A **category** is —

- **A.** a sudden flash of light.
- **B.** a group of things that are similar to each other in some way.
- **C.** a tool for measuring rain.
- **D.** a path a storm follows.

Q31 L.5.4

track

Scientists can **track** hurricanes better today. To **track** something is to —

- **A.** push it with great force.
- **B.** hide it from view.
- **C.** watch it and see where it moves and how it changes.
- **D.** repair it when it breaks.

Worksheet B — Reading Comprehension

Module 3 · Natural Disasters · Grade 5 · 32 Items

CCSS RI.5.2 · RI.5.3 · RI.5.5 · RI.5.6 · RI.5.8

NAME _____ DATE _____

Directions. Use what you remember from each selection to answer the questions. Fill in the circle next to the BEST answer.

Selection 1 · Eruption! Volcanoes and the Science of Saving Lives

NARRATIVE NONFICTION · ELIZABETH RUSCH · 8 QUESTIONS

Q1 RI.5.2

What is "Eruption!" mostly about?

- ◯ **A.** A town that learns to rebuild after a tsunami.
- ◯ **B.** The story of a single earthquake in Alaska.
- ◯ **C.** A team of volcano scientists who monitor Mount Pinatubo and persuade people to evacuate before it erupts, saving many lives.
- ◯ **D.** How hurricanes form over warm ocean water.

Q2 RI.5.3

Why did Andy Lockhart join a volcano crisis team?

- ◯ **A.** He was horrified by an earlier eruption that killed many people and wanted to help prevent such tragedies.
- ◯ **B.** He wanted to become a famous photographer.
- ◯ **C.** He hoped to live near a volcano for fun.
- ◯ **D.** He was told to do so by the military.

Q3 RI.5.8

Why was the scientists' decision about whether to call for an evacuation so difficult?

- ◯ **A.** They were sure nothing would happen.
- ◯ **B.** They did not have any instruments to use.
- ◯ **C.** They were not allowed to talk to one another.
- ◯ **D.** A false alarm would be costly, but waiting too long could cause a tragic loss of life, so the consequences weighed heavily on them.

Q4 RI.5.3

How did the seismographs help the scientists?

- **A.** They carried people away from the volcano.
- **B.** They recorded the earthquakes so the scientists could tell that pressure was building inside the volcano.
- **C.** They cooled down the hot magma.
- **D.** They blocked the ash from falling.

Q5 RI.5.5

How does the author organize the events of this selection?

- **A.** As a list of rhyming poems.
- **B.** As a set of letters between two friends.
- **C.** By comparing two different countries side by side.
- **D.** In chronological order, telling a true story as the events happened over time.

Q6 RI.5.6

Why does the author include the scientists' own words and the photographs?

- **A.** To teach readers how to draw a volcano.
- **B.** To explain how to build an airplane.
- **C.** To help readers understand the real events and feel the danger the scientists faced.
- **D.** To prove that the events never really happened.

Q7 RI.5.3

What happened after the scientists conferred with one another and warned the military?

- **A.** People were moved away from the danger, and when Mount Pinatubo erupted, many lives were saved.
- **B.** The volcano stopped making any noise.
- **C.** Nothing changed at the air base.
- **D.** The scientists left the country at once.

Q8 RI.5.2

Which sentence BEST states a central idea of "Eruption!"?

- **A.** Scientists should never study dangerous places.
- **B.** By carefully monitoring a volcano and making hard decisions, scientists can predict an eruption and save thousands of lives.
- **C.** Volcanoes are found only in the Philippines.
- **D.** Earthquakes are always harmless.

Selection 2 · Between the Glacier and the Sea: The Alaska Earthquake

MEDIA: VIDEO · FROM THE VALDEZ MUSEUM · 8 QUESTIONS

Q9 RI.5.2

What is this video mostly about?

- ◯ **A.** How a team of scientists watched a volcano.
- ◯ **B.** The 1964 Great Alaska Earthquake and how it affected the town of Valdez.
- ◯ **C.** How coral can show the dates of past earthquakes.
- ◯ **D.** How hurricanes are named and ranked.

Q10 RI.5.6

Why does the video include first-person accounts from people who lived through the earthquake?

- ◯ **A.** To teach viewers how to build oil tanks.
- ◯ **B.** To list the names of every street in Valdez.
- ◯ **C.** To prove the earthquake never happened.
- ◯ **D.** To help viewers better understand what the event was really like for the people who survived it.

Q11 RI.5.5

How do the narration and the animations work together in this video?

- ◯ **A.** The narrator explains what is happening while the animations show how the earth's plates moved, so viewers can both hear and see the ideas.
- ◯ **B.** The animations replace the need for any sound.
- ◯ **C.** The animations show only colors and shapes with no meaning.
- ◯ **D.** The narration and animations tell two different stories.

Q12 RI.5.2

Which detail BEST supports the idea that this earthquake was extreme?

- ◯ **A.** The shaking lasted only a few seconds.
- ◯ **B.** A few windows were cracked in one home.
- ◯ **C.** Its 9.2 magnitude was the largest ever recorded in North American history, and the ground shook violently for about five minutes.
- ◯ **D.** The town stayed exactly the same afterward.

Q13 RI.5.3

According to the video, what happened to the town of Valdez during the disaster?

- ◯ **A.** The town was left completely untouched.
- ◯ **B.** The town grew larger during the shaking.
- ◯ **C.** Much of the town was destroyed, and the earthquake led to dangerous fires.
- ◯ **D.** The town was moved before the earthquake began.

Q14 RI.5.3

What does the word *magnitude* help viewers understand about the earthquake?

- ◯ **A.** How strong, or how great in size, the earthquake was.
- ◯ **B.** The color of the sky during the quake.
- ◯ **C.** How long the maintenance of the oil tanks took.
- ◯ **D.** How many people lived in the town.

Q15 RI.5.6

What is the main purpose of this video?

- ◯ **A.** To sell tickets to a museum.
- ◯ **B.** To inform viewers about the causes and human impact of the 1964 Alaska earthquake.
- ◯ **C.** To teach viewers how to drive a boat.
- ◯ **D.** To make viewers laugh at a funny story.

Q16 RI.5.2

Which sentence BEST states a central idea of this video?

- ◯ **A.** Oil tanks should be painted gray.
- ◯ **B.** Earthquakes only happen in small towns.
- ◯ **C.** Alaska has very cold winters.
- ◯ **D.** Firsthand accounts and visuals together help people understand the power and damage of a major earthquake.

Selection 3 · Quaking Earth, Racing Waves

INFORMATIONAL TEXT · RACHEL YOUNG · 8 QUESTIONS

Q17 RI.5.2

What is "Quaking Earth, Racing Waves" mostly about?

- ◯ **A.** How a team of scientists watched a volcano in the Philippines.
- ◯ **B.** The story of a single earthquake in Alaska.
- ◯ **C.** How hurricanes are ranked from Category 1 to Category 5.
- ◯ **D.** How earthquakes and tsunamis form, and how a scientist used clues to predict one.

Q18 RI.5.5

How is this informational text organized?

- **A.** As a play with scenes and dialogue.
- **B.** Under subheadings, with diagrams and a map that tell more about the topic.
- **C.** As one long poem.
- **D.** As a series of dated journal entries.

Q19 RI.5.3

How did Kerry Sieh figure out that a major earthquake was likely near the island?

- **A.** He asked the villagers what they remembered.
- **B.** He measured the height of the nearby mountains.
- **C.** He studied the growth patterns of coral heads near the fault to learn the dates of past earthquakes.
- **D.** He guessed by looking at the clouds.

Q20 RI.5.3

According to the text, how is a tsunami started?

- **A.** By a disturbance, such as an undersea earthquake, that shifts water at the ocean floor.
- **B.** By strong winds blowing across the water, like most ordinary waves.
- **C.** By rain falling into the ocean.
- **D.** By the heat of the summer sun.

Q21 RI.5.8

What evidence does the author give to support the idea that earthquakes near the island happen in a pattern?

- **A.** The coral growth records showed that earthquakes occurred about every 200 years.
- **B.** The ocean was always perfectly calm.
- **C.** No earthquake had ever happened there before.
- **D.** The villagers never felt any shaking at all.

Q22 RI.5.3

Why is the town described as safer after it was modified?

- **A.** Because the town was moved out to sea.
- **B.** Because the earthquakes had stopped forever.
- **C.** Because the changes made the town better prepared for tsunamis.
- **D.** Because the people decided to stop watching the ocean.

Q23 RI.5.6

Why does the author include diagrams and a map in this text?

- **A.** To replace the need for any words.
- **B.** To make the text longer.
- **C.** To show how to draw a boat.
- **D.** To help readers picture how the earth moves and where the events take place.

Q24 RI.5.2

Which sentence BEST states a central idea of "Quaking Earth, Racing Waves"?

- **A.** Islands are always dangerous places to live.
- **B.** By studying evidence such as corals and plate movement, scientists can predict earthquakes and tsunamis and help people prepare.
- **C.** Coral is the prettiest thing in the ocean.
- **D.** Waves are all formed in exactly the same way.

Selection 4 · Hurricanes: The Science Behind Killer Storms

INFORMATIONAL TEXT · ALVIN & VIRGINIA SILVERSTEIN AND LAURA SILVERSTEIN NUNN · 8 QUESTIONS

Q25 RI.5.2

What is "Hurricanes" mostly about?

- **A.** How a team of scientists watched a volcano erupt.
- **B.** How hurricanes form, what dangers they bring, and how people can stay safe.
- **C.** The story of one earthquake in Alaska.
- **D.** How coral can show the dates of past earthquakes.

Q26 RI.5.5

How does the author help readers find the central ideas in this text?

- **A.** By telling the story through rhyming poems.
- **B.** By writing the whole text as one long paragraph.
- **C.** By using dialogue and stage directions.
- **D.** By using headings, along with charts, a map, and sidebar features that tell more about the topic.

Q27 RI.5.3

According to the text, what are the four stages a hurricane goes through as it grows?

- **A.** Eye, eyewall, rainband, and surge.
- **B.** Category 1, 2, 3, and 4.
- **C.** Tropical disturbance, tropical depression, tropical storm, and hurricane.
- **D.** Spring, summer, fall, and winter.

Q28 RI.5.8

What reason does the text give to explain why mobile homes are especially dangerous in a hurricane?

- **A.** They are not well anchored to the ground, so strong winds can move or damage them easily.
- **B.** They are always painted bright colors.
- **C.** They are built deep underground.
- **D.** They are too heavy for the wind to move.

Q29 RI.5.3

Why are people more likely to die from a storm surge than from a hurricane's winds?

- **A.** A storm surge makes the air very cold.
- **B.** A storm surge is only a light sprinkle of rain.
- **C.** A storm surge can push a huge, sudden rush of seawater onto the land.
- **D.** A storm surge blocks out the sunlight.

Q30 RI.5.3

How does the Saffir-Simpson scale help people during hurricane season?

- **A.** It ranks hurricanes into categories so people know how strong and dangerous a storm is.
- **B.** It tells people what to wear each day.
- **C.** It names the streets in a town.
- **D.** It measures how much rain falls in a year.

Q31 RI.5.6

What is the authors' main purpose in writing "Hurricanes"?

- **A.** To convince readers that hurricanes are not real.
- **B.** To tell a make-believe adventure story.
- **C.** To describe how to paint a house.
- **D.** To explain the science of hurricanes so readers understand the danger and can prepare.

Q32 RI.5.2 TRAP

Which sentence BEST states the central idea of "Hurricanes"?

- ◯ **A.** Some hurricanes are given names.
- ◯ **B.** Understanding how hurricanes form and what damage they cause helps people prepare and stay safe.
- ◯ **C.** A hurricane has an eye at its center.
- ◯ **D.** A hurricane scale goes up to Category 5.

Worksheet C — IB Critical Thinking

Module 3 · Natural Disasters · Grade 5

IB PYP Key Concepts · DOK 2 – 4

NAME _____ DATE _____

Directions. Use what you learned from each selection to think more deeply about how science helps us understand and survive natural disasters. Each question helps you grow as an IB learner.

Q1 Causation DOK 3

In **"Eruption!,"** scientists watched for warning signs before Mount Pinatubo erupted. Show the CAUSE and the EFFECT, then explain how their work helped people.

CAUSE AND EFFECT · ERUPTION!

WHAT THE SCIENTISTS NOTICED	→	WHAT THEY DECIDED TO DO

Why was the scientists' decision so hard? Use the word *consequences* in your answer.

Q2 Perspective DOK 3

In **"Between the Glacier and the Sea,"** the video uses different tools to tell about the 1964 Alaska earthquake. Show TWO tools it uses, then explain why firsthand accounts matter.

HOW A VIDEO TELLS THE STORY · BETWEEN THE GLACIER AND THE SEA

A TOOL THE VIDEO USES	ANOTHER TOOL THE VIDEO USES

Why does hearing from people who lived through the earthquake help you understand it better than facts alone? Explain.

Q3 Systems DOK 3

In **"Quaking Earth, Racing Waves,"** an earthquake under the sea can lead to a tsunami. Show the STEPS, then explain how a scientist predicted the danger.

FROM QUAKE TO WAVE · QUAKING EARTH, RACING WAVES

THE CAUSE	WHAT THE WATER DOES	THE RESULT
what happens under the sea?	*how does the wave form?*	*what reaches the shore?*

Kerry Sieh read clues in the corals to warn people. Why is it important that he did not ignore the evidence? Explain using the word *triggered*.

Q4 Responsibility DOK 3

In **"Hurricanes,"** the authors explain the dangers of storms and how people can stay safe. Show one DANGER and one way to PREPARE, then explain why this knowledge matters.

DANGER AND PREPARATION · HURRICANES

A DANGER A HURRICANE BRINGS	A WAY PEOPLE CAN STAY SAFE

The authors wrote this text to explain the science of hurricanes. Why does understanding how a storm works help people prepare? Use the word *category* in your answer.

Q5 Inquiry DOK 4

Essential Question: How can learning about natural disasters make us safer? Use ideas from TWO of the selections in this module to answer.

TWO-SELECTION SYNTHESIS

SELECTION 1 · TITLE

What disaster does it explain, and how does learning about it help keep people safe?

SELECTION 2 · TITLE

What disaster does it explain, and how does learning about it help keep people safe?

MY BIG IDEA

Worksheet D — MAP Practice

Module 3 · Natural Disasters · Grade 5

MAP RIT 205 → 225 · 4 Mini-Passages × 5 Questions

NAME _____ DATE _____

Directions. Read each short passage carefully. Then answer the 5 questions that follow. Each passage is NEW — not from the selections in this workbook.

Passage 1 · The Mountain That Smoked

On a quiet island in the sea, a tall mountain rose above a small village. For as long as anyone could remember, the mountain had been still. But it was not an ordinary mountain. It was a volcano, and deep inside it, melted rock was slowly building up. One spring, the ground near the village began to shake with small tremors. A thin trail of steam rose from the top of the mountain, and a faint smell of rotten eggs drifted through the streets. A scientist named Dr. Reyes came to study the volcano. She set up instruments that measured the tiny earthquakes, and she watched the steam grow thicker each day. The signs told her that pressure was building inside the mountain. Dr. Reyes warned the village leaders that the volcano could erupt soon. At first, some people did not want to leave their homes. But the leaders trusted the science, and they moved everyone to a safer town farther away. Two weeks later, the mountain roared. Ash and rock burst into the sky, and hot mud poured down the slopes. The village was buried, but not a single person was hurt. The warning signs had saved them all.

200 WORDS · LEXILE 640–720L · RIT 205–211

Q1 RIT 205 | Lexile 640L | Key Detail

What kind of mountain rose above the village?

- ◯ **A.** A hill covered with green grass.
- ◯ **B.** A pile of snow and ice.
- ◯ **C.** A volcano with melted rock building up inside it.
- ◯ **D.** A mountain made of sand.

Q2 RIT 207 | Lexile 660L | Cause & Effect

Why did Dr. Reyes warn the village leaders?

- ◯ **A.** The villagers had asked her to leave.
- ◯ **B.** The warning signs told her the volcano could erupt soon.
- ◯ **C.** She wanted to build a new town.
- ◯ **D.** She was afraid of the rain.

Q3 RIT 208 · Lexile 680L · Key Detail

What did Dr. Reyes do to study the volcano?

- ◯ **A.** She set up instruments that measured the tiny earthquakes and watched the steam.
- ◯ **B.** She climbed inside the volcano.
- ◯ **C.** She painted the mountain.
- ◯ **D.** She moved the village by herself.

Q4 RIT 210 · Lexile 700L · Vocabulary

The passage says the ground shook with small **tremors**. The word **tremors** means —

- ◯ **A.** bright lights.
- ◯ **B.** loud songs.
- ◯ **C.** cold winds.
- ◯ **D.** small shaking movements of the ground.

Q5 RIT 211 · Lexile 720L · Main Idea

What is the main idea of this passage?

- ◯ **A.** Islands are the best places to live.
- ◯ **B.** Reading the warning signs of a volcano can help people stay safe.
- ◯ **C.** Mountains should never be studied.
- ◯ **D.** Steam always smells like rotten eggs.

Passage 2 · What Makes the Ground Shake?

Have you ever wondered why the ground sometimes shakes during an earthquake? The answer lies far beneath your feet. The outer shell of the Earth is not one solid piece. Instead, it is broken into giant slabs of rock called plates. These plates fit together like the pieces of a cracked eggshell, and they are always moving — but so slowly that we cannot feel it. Sometimes two plates push against each other along a line called a fault. The rough edges of the plates can get stuck. Pressure builds up for many years, the way it builds when you bend a stick further and further. Finally, the rock can hold no more, and the plates suddenly slip. All that stored energy is released at once, and it radiates outward in waves. Those waves make the ground shake. Scientists use a special tool to record the size, or magnitude, of each earthquake. A small earthquake may rattle the dishes, while a large one can crack roads and topple buildings. By studying faults and measuring quakes, scientists learn which areas are most likely to shake. This knowledge helps engineers design safer buildings and helps families know how to prepare before the ground starts to move.

203 WORDS · LEXILE 720–820L · RIT 211–217

Q6 RIT 211 · Lexile 720L · Key Detail

According to the passage, what is the outer shell of the Earth made of?

- **A.** One solid piece of rock.
- **B.** Soft layers of sand and water.
- **C.** Thick sheets of ice.
- **D.** Giant slabs of rock called plates that fit together like a cracked eggshell.

Q7 RIT 213 · Lexile 750L · Cause & Effect

What happens when two stuck plates finally slip?

- **A.** Stored energy is released at once and radiates outward in waves that make the ground shake.
- **B.** The plates melt into liquid.
- **C.** The Earth stops moving forever.
- **D.** The plates float up into the sky.

Q8 RIT 214 · Lexile 780L · Text Structure

How does the author mainly organize this passage?

- **A.** By telling a make-believe story about a hero.
- **B.** By listing the days of the week in order.
- **C.** By comparing two different towns.
- **D.** By explaining the causes that lead to the effect of an earthquake.

Q9 RIT 215 · Lexile 800L · Vocabulary

In this passage, the word **magnitude** means the —

- **A.** color of the rocks below ground.
- **B.** name of the nearest town.
- **C.** size, or strength, of an earthquake.
- **D.** number of plates on Earth.

Q10 RIT 217 · Lexile 820L · Main Idea

What is the main idea of "What Makes the Ground Shake?"

- **A.** Moving plates cause earthquakes, and studying them helps people build safer buildings and prepare.
- **B.** Eggshells are easy to crack.
- **C.** Sticks bend when you push them.
- **D.** Dishes rattle for no reason at all.

Passage 3 · The Wave That Saved a Town

Maya lived in a fishing town on the coast, where her grandmother had taught her an old lesson: if the sea ever pulls far away from the shore, run for high ground. One warm morning, the earth gave a long, rolling shake. Pots fell from shelves, and the dogs began to bark. When the shaking stopped, the people gathered on the beach to look at the damage. Then Maya noticed something strange. The ocean was sliding backward, away from the land, leaving fish flapping on the wet sand. Boats tipped over in the mud where water had been moments before. Maya's heart pounded. She remembered her grandmother's words. "The water is leaving!" she shouted. "A big wave is coming! Run uphill, everyone, run!" Some people stared, unsure. But Maya did not wait. She grabbed her little brother's hand and raced up the steep path toward the hills, calling for the others to follow. One by one, they did. Minutes later, a towering wall of seawater rushed in and swallowed the beach where they had all been standing. From the safety of the hill, the people watched in silence. A child's memory of an old warning, passed down through her family, had saved the whole town.

205 WORDS · LEXILE 760–860L · RIT 217–222

Q11 RIT 217 Lexile 760L Key Detail

What lesson had Maya's grandmother taught her?

- ◯ **A.** Always go fishing in the morning.
- ◯ **B.** Never go near the dogs.
- ◯ **C.** If the sea pulls far away from the shore, run for high ground.
- ◯ **D.** Keep the pots on low shelves.

Q12 RIT 218 Lexile 790L Cause & Effect

Why did Maya start shouting for everyone to run uphill?

- ◯ **A.** She wanted to win a race.
- ◯ **B.** She saw the ocean sliding backward and remembered that it meant a big wave was coming.
- ◯ **C.** She saw a storm in the sky.
- ◯ **D.** Her brother told her to.

Q13 RIT 219 Lexile 820L Inference

Why did some people stare and feel unsure at first?

- ◯ **A.** They were too tired to move.
- ◯ **B.** They did not understand what the strange sight meant.
- ◯ **C.** They wanted to catch the flapping fish.
- ◯ **D.** They could not hear Maya at all.

Q14 RIT 220 · Lexile 840L · Cause & Effect

What happened after the people followed Maya up the hill?

- ◯ **A.** A towering wall of seawater rushed in and swallowed the beach where they had been standing.
- ◯ **B.** The sea returned to normal right away.
- ◯ **C.** The dogs ran into the water.
- ◯ **D.** The boats floated back to shore.

Q15 RIT 222 · Lexile 860L · Theme

What is the lesson, or theme, of this passage?

- ◯ **A.** Knowing the warning signs of danger can save lives.
- ◯ **B.** Fishing towns are dangerous places.
- ◯ **C.** Dogs always know when danger is near.
- ◯ **D.** It is best to stay on the beach during an earthquake.

Passage 4 · How a Hurricane Is Born

A hurricane is one of the most powerful storms on Earth, yet it begins in a gentle way. It starts over warm ocean water near the equator, where the sun heats the sea all year. As the warm water evaporates, it rises into the air and forms clouds. More warm, moist air rushes in to take its place, and it rises too. Slowly, the rising air begins to spin. As it spins faster, it pulls in more and more moisture, and the storm grows wider and stronger. When the winds reach a certain speed, the storm is officially called a hurricane. At the very center is a calm, clear area called the eye. Around the eye spins a ring of the fiercest winds and heaviest rain. A hurricane can stretch for hundreds of miles, and it moves slowly across the ocean, gathering strength as long as it stays over warm water. But once it reaches land or drifts over cooler seas, it loses its supply of warm, moist air, and it begins to weaken. Today, scientists use satellites high above the Earth to watch hurricanes and follow their path. These tools give people days of warning, so families near the coast have time to get ready.

206 WORDS · LEXILE 820–940L · RIT 219–225

Q16 RIT 219 · Lexile 820L · Key Detail

Where does a hurricane begin?

- ◯ **A.** High on a snowy mountain.
- ◯ **B.** Deep underground.
- ◯ **C.** In the middle of a desert.
- ◯ **D.** Over warm ocean water near the equator.

Q17 RIT 220 | Lexile 850L | Cause & Effect

Why does a hurricane begin to weaken when it reaches land?

- **A.** It spins in the opposite direction.
- **B.** It grows a second eye.
- **C.** It loses its supply of warm, moist air.
- **D.** The sun stops shining on it.

Q18 RIT 221 | Lexile 880L | Key Detail

What is the calm, clear area at the center of a hurricane called?

- **A.** The rainband.
- **B.** The surge.
- **C.** The fault.
- **D.** The eye.

Q19 RIT 223 | Lexile 910L | Vocabulary

The passage says warm water **evaporates** and rises into the air. The word **evaporates** means —

- **A.** freezes into hard ice.
- **B.** changes into a vapor and rises into the air.
- **C.** sinks to the bottom.
- **D.** turns a dark color.

Q20 RIT 225 | Lexile 940L | Main Idea

What is the main idea of "How a Hurricane Is Born"?

- **A.** Hurricanes are too small to be seen.
- **B.** Satellites are only used to study the moon.
- **C.** A hurricane forms over warm water and grows stronger there, and scientists track it to warn people in time.
- **D.** The eye of a storm is the most dangerous part.